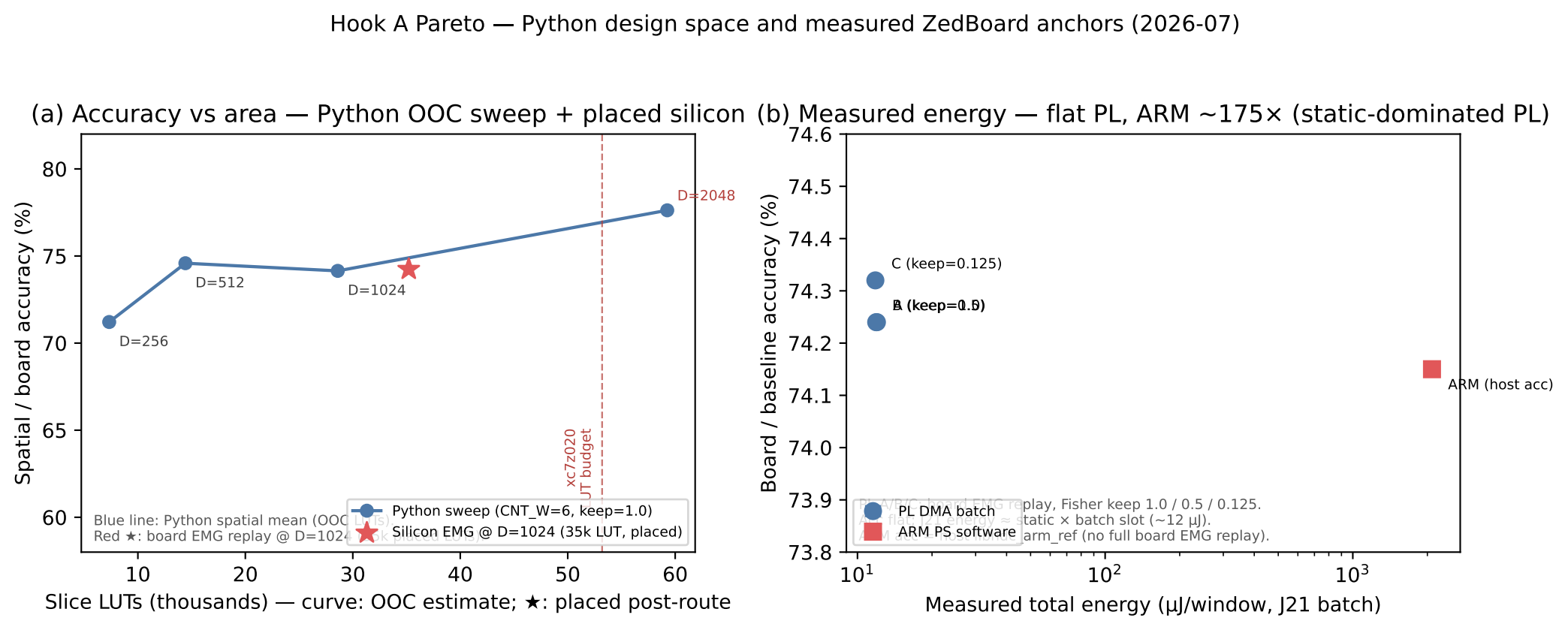
Hook A Pareto — Python design space and measured ZedBoard anchors (2026-07)
(a) Accuracy vs area — Python OOC sweep + placed silicon
(b) Measured energy — flat PL, ARM ~175× (static-dominated PL)
Spatial / board accuracy (%)
Slice LUTs (thousands) — curve: OOC estimate; ★: placed post-route
D=256
D=512
D=1024
D=2048
xc7z020 LUT budget
Python sweep (CNT_W=6, keep=1.0)
Silicon EMG @ D=1024 (35k LUT, placed)
Blue line: Python spatial mean (OOC LUTs)
Red ★: board EMG replay @ D=1024 (35k placed LUTs)
Board / baseline accuracy (%)
Measured total energy (µJ/window, J21 batch)
C (keep=0.125)
B (keep=0.5)
A (keep=1.0)
ARM (host acc)
PL DMA batch
ARM PS software
PL A/B/C: board EMG replay, Fisher keep 1.0 / 0.5 / 0.125.
PL flat J21 energy ≈ static × batch slot (~12 µJ).
ARM acc = host arm_ref (no full board EMG replay).